INFRASTRUCTURE PLANNING REPORT

Vida Hosting & Scaling Cost Analysis

Render, MongoDB Atlas, Cloudflare R2, Android distribution and domain costs

$16-$18/month beta baseline → **$710-$902/month** at 100,000 MAU

Prepared for Vida

20 July 2026 | Singapore

Executive summary

PLANNING RECOMMENDATION **Budget about $20/month for a controlled beta, $85/month for a dependable small production launch, $110/month near 10,000 MAU, and approximately $800/month near 100,000 MAU.**

The number of registered accounts alone does not drive hosting cost. This report uses monthly active users (MAU), because requests, bandwidth, stored content and peak concurrency are what create spend.

Cost outlook at a glance

Monthly active users	Estimated monthly	Estimated annual	Practical interpretation
100	$16-$18	$192-$216	Lean beta. Shared database and one small API instance.
1,000	$17-$45	$204-$540	Controlled beta or early launch. Upgrade if uptime is business-critical.
10,000	$103-$114	$1,236-$1,368	Standard API plus dedicated Atlas M10.
100,000	$710-$902	$8,520-$10,824	Pro workspace, one or two Pro API instances and Atlas M30.

Table 1. Core infrastructure estimate; USD before tax. Optional maps and third-party messaging are excluded.

What matters most

- **Media architecture is cost-efficient.** Images upload and download through Cloudflare R2 rather than being proxied through Render.
- **Database reliability creates the first meaningful fixed-cost step.** Atlas Flex is inexpensive, but MongoDB positions it for development and testing; an M10 is the more credible production floor.
- **Bandwidth becomes material at scale.** Under Render's new Hobby allowance, 100 GB/month costs about $14 in overage and 1 TB/month costs roughly $146 on Pro.
- **Map services are the main unpriced dependency.** The current public OpenStreetMap tile and Photon endpoints provide no commercial SLA and can throttle heavy use.

Detailed monthly cost breakdown

The figures below represent a reasonable capacity plan for the current architecture. They are budget estimates rather than vendor quotes; actual spend varies with region, traffic shape, query efficiency and retention.

Cost component	100 MAU	1,000 MAU	10,000 MAU	100,000 MAU
Render API compute	$7	$7	$25	$85-$170
Render workspace plan	$0	$0	$0	$25
Render static frontends	$0	$0	$0	$0
Render bandwidth	$0	~$0.75	~$14	~$146
MongoDB Atlas	$8	$8-$30	~$57	~$394
Atlas backup / transfer reserve	$0	$0-$5	$5-$15	$50-$150
Cloudflare R2 media	$0	$0	~$1	$9-$15
Domain registration	$1-$2	$1-$2	$1-$2	$1-$2
Estimated total	**$16-$18**	**$17-$45**	**$103-$114**	**$710-$902**

Table 2. Estimated recurring monthly spend. The Google Play developer registration fee is a separate one-time cost.

Why each component scales this way

Render static frontends. Both Vite frontends can remain on Render's $0 static-site tier. They still consume the workspace's bandwidth and build-minute allowances.

Render API. The Express service starts comfortably on Starter for a beta, moves to Standard around 10,000 active users, and uses one or two Pro instances near 100,000 MAU for capacity and redundancy.

MongoDB Atlas. Flex covers low-volume development workloads up to 5 GB. M10 provides a 10 GB dedicated starting point; M30 provides 40 GB and more predictable performance for higher traffic.

Cloudflare R2. The generous free tier and zero internet-egress fee keep image costs low. At 100,000 MAU, stored media is more important than delivery bandwidth.

PRODUCTION FLOOR **If uptime matters from day one, use Render Standard plus Atlas M10 even at 100-1,000 MAU. That produces a safer baseline of roughly $83-$86 per month.**

Workload assumptions

All estimates depend on a concrete traffic model. This report uses the following middle-of-the-road assumptions for Vida's social, activity and group features:

- 300 API calls per active user per month.
- Approximately 10 MB of Render-delivered data per active user per month; R2-hosted images are excluded from this figure.
- Approximately 0.5 MB of newly retained image data per active user per month, retained for twelve months.
- Peak request rates of approximately ten times the monthly average.
- MongoDB and Render deployed on the same cloud provider and region where possible.
- Efficient pagination, indexing and field projection remain in place as the dataset grows.

Recommended scaling path

1. **Beta: up to about 1,000 MAU:** Render Starter + Atlas Flex + free static sites + R2. Budget $20-$45/month and accept shared-tier limitations.
2. **Dependable launch:** Move to Render Standard + Atlas M10. Budget approximately $85/month even if the user count is still small.
3. **Growth: around 10,000 MAU:** Keep one Standard API instance initially, use Atlas M10, monitor p95 latency, memory and slow queries, and budget about $110/month.
4. **Scale: around 100,000 MAU:** Use Render Pro workspace features, one or two Pro instances, Atlas M30 with autoscaling, and budget roughly $800/month before map services.

Architecture observations

The repository currently contains one Express/Mongoose backend, two React/Vite web frontends and a native Android client. Authentication is token-based, and the backend does not rely on local persistent storage. This makes horizontal API scaling straightforward.

Image uploads use short-lived presigned URLs, so the client transfers files directly to R2. This avoids paying Render bandwidth for large media uploads and prevents the API server from becoming a file-transfer bottleneck.

MongoDB transactions are used for participation and attendance operations. This supports consistency, but it also makes database health and correct regional placement more important than raw account count.

Risk, distribution and cost sensitivities

Sensitivity rules

- **Each additional 1 MB of Render egress per MAU** adds about $15/month at 100,000 MAU after the allowance is exhausted.
- **If only 20% of registered accounts are active,** usage-driven bandwidth and operation costs are closer to one-fifth of the table assumptions; fixed compute costs do not fall proportionally.
- **At 100,000 MAU,** Render Pro plus bandwidth overage is still cheaper than choosing the $499 Scale workspace solely for its 1 TB allowance.

Android distribution and domain

Google Play distribution. A developer account costs $25 once. A free application does not incur a per-install hosting fee. If Vida later sells digital goods or subscriptions, Google Play service fees may apply as a percentage of transaction value.

Domain registration. Plan approximately $10-$20 per year for a common domain extension. Render includes managed TLS certificates. One root domain can serve app, vendor and API subdomains, although Render may charge $0.25/month for custom hostnames beyond the workspace allowance.

Potentially significant map cost

ACTION REQUIRED BEFORE SCALE **Replace or contractually validate the public OpenStreetMap tile and Photon demo endpoints before large-scale production use.**

Vida's web code currently loads OpenStreetMap community tiles and uses Photon for location search. OpenStreetMap states that its tile service has limited capacity and no SLA, while Photon warns that extensive demo-server usage may be throttled or banned.

A commercial map and geocoding provider could add approximately $0 at low traffic, $0-$100/month near 10,000 MAU, and $100-$1,000+ near 100,000 MAU. This range is excluded from the core totals because map-open and search frequency are not yet measured.

Items outside the core estimate

Not included in the core estimate

- Email delivery, password-reset email and transactional messaging.
- SMS or WhatsApp notifications.
- Commercial monitoring, error tracking, log retention and incident response.
- Paid analytics, customer support platforms or moderation tooling.
- Google Play revenue-share fees for paid digital products.
- Engineering, security review, privacy compliance and support staff.
- Taxes, currency conversion and vendor-specific regional premiums.

Immediate planning actions

1. Treat the first 1,000 MAU as an instrumentation phase and measure API egress per user, requests per session, database size and R2 storage growth.
2. Set billing alerts in Render, Atlas and Cloudflare before public launch.
3. Place Render and Atlas in matching AWS regions to minimize latency and transfer cost.
4. Upgrade to the $85/month production floor when downtime begins to affect real users, partners or revenue.
5. Review mapping infrastructure before marketing-driven growth or Android release at scale.

Sources and methodology

Pricing was checked against official vendor documentation on 20 July 2026. Vendor rates can change, and MongoDB pricing varies by cloud provider and region. All calculations are rounded for planning use.

1. Render pricing - Compute tiers, static sites, workspace plans and bandwidth overage.
2. Render new workspace plans - The Hobby bandwidth allowance changes from 100 GB to 5 GB under the new plan; remaining legacy workspaces migrate on 1 August 2026.
3. MongoDB Atlas pricing - Flex and dedicated cluster starting prices and resource tiers.
4. Atlas Flex costs - The $8-$30 monthly range, 5 GB storage and operation limits.
5. Atlas data-transfer costs - Region and provider effects on transfer charges.
6. Cloudflare R2 pricing - Storage, operation free tiers and zero internet-egress fee.
7. Google Play developer registration - The one-time $25 developer registration fee.
8. Cloudflare Registrar - At-cost domain registration and renewal.
9. OpenStreetMap tile usage policy - Capacity limits, best-effort availability and production-use requirements.
10. Photon geocoder project - Public demo endpoint and throttling warning.

Repository basis

Backend overview: backend/README.md

Express entry point: backend/src/server.ts

R2 integration: backend/src/lib/r2.ts

User web frontend: frontend/

Vendor web frontend: vendor-frontend/

Native Android client: user-frontend/

USE OF THIS REPORT **This is a capacity-planning estimate, not a contractual quote. Reforecast after the first month of real traffic and again before each major growth campaign.**